Table S-ADMET. ADMETlab 3.0 predicted ADMET endpoints for the 15 core metabolites (descriptive only)

Metabolite	Caco-2 (log Papp)	P(HIA < 30%)	P(Pgp inhibitor)	P(Pgp substrate)	PPB (%)	Fu (unbound, %)	P(BBB+)	P(CYP1A2 inh.)	P(CYP2C9 inh.)	P(CYP2C19 inh.)	P(CYP2D6 inh.)	P(CYP3A4 inh.)	P(hERG blocker)	P(DILI)	P(Ames +)	P(carcinogen)
Acacetin	-4.843	0.0	0.025	0.007	97.808	2.466	0.205	0.901	0.573	0.644	0.251	0.604	0.194	1.0	0.777	0.533
Apigenin	-4.984	0.0	0.0	0.025	99.149	1.132	0.001	0.834	0.53	0.532	0.236	0.517	0.804	1.0	0.734	0.063
Dopamine	-5.019	0.981	0.0	0.375	10.39	85.114	0.008	0.133	0.028	0.08	0.201	0.033	0.001	0.0	0.963	0.048
Genistein	-4.942	0.0	0.0	0.001	98.013	1.811	0.0	0.773	0.509	0.49	0.212	0.469	0.881	1.0	0.444	0.06
Levodopa	-5.909	0.0	0.0	0.999	21.414	89.773	0.001	0.029	0.016	0.054	0.048	0.007	0.0	0.0	0.968	0.018
Lithocholic Acid	-5.37	0.0	0.991	0.0	94.154	6.254	0.025	0.063	0.104	0.183	0.147	0.086	0.012	0.012	0.013	0.998
Quercetin	-6.249	0.004	0.0	0.0	99.435	0.72	0.0	0.663	0.396	0.334	0.152	0.355	0.979	1.0	0.984	0.036
5-(3,4-Dihydroxyphenyl)pentanoic acid	-4.843	0.248	0.0	0.002	74.053	16.852	0.008	0.097	0.069	0.154	0.08	0.019	0.003	0.029	0.596	0.026
5-(3-Hydroxyphenyl)pentanoic Acid	-4.516	0.022	0.0	0.011	75.644	15.318	0.01	0.165	0.099	0.197	0.102	0.03	0.005	0.093	0.423	0.027
Acetate	-5.092	0.002	0.0	0.058	26.515	53.892	0.497	0.01	0.006	0.039	0.021	0.001	0.001	0.023	0.047	0.347
Butyrate	-4.326	0.001	0.0	0.003	43.549	44.678	0.993	0.015	0.007	0.06	0.024	0.001	0.001	0.87	0.001	0.762
Equol	-4.881	0.0	0.001	0.053	89.199	15.2	0.913	0.866	0.575	0.71	0.494	0.514	0.254	0.881	0.687	0.866
Kaempferol	-5.845	0.0	0.048	0.0	98.816	0.866	0.0	0.725	0.458	0.414	0.191	0.417	0.987	1.0	0.951	0.089
Luteolin	-5.301	0.004	0.0	0.008	98.99	1.274	0.0	0.758	0.466	0.436	0.2	0.461	0.969	1.0	0.956	0.098
Myricetin	-6.557	0.004	0.001	0.0	99.039	1.02	0.0	0.606	0.37	0.272	0.126	0.31	0.963	1.0	0.977	0.001

Predictions from the ADMETlab 3.0 web service (admetlab3.scbdd.com, API call made on 2026-10-25) using the standardised SMILES of Table S-physchem; model outputs, not measurements. P(...) = model probability of the stated class. Note the direction of the absorption column: ADMETlab 3.0 reports the probability of HIA+, defined as human intestinal absorption < 30%, so a small value means good predicted absorption. Caco-2 is log Papp (cm/s; the service flags values above -5.15 as optimal). PPB and Fu are model predictions (both in %; Fu is the predicted fraction unbound in plasma, low < 5%), not binding measurements, and are not used in the concentration chain (E63/E67), which uses literature fu only. SwissADME was not run: its web form cannot be submitted from a script. The table is descriptive and was not used to select metabolites. Gut-derived metabolites are largely endogenous or microbial products and can act locally in the gut lumen; drug-likeness and toxicity classifiers trained on drugs are given for reference only. All 15 compounds returned a result. AI preliminary calculation (E65).